www.tntextbooks.in

Its gravitational force ($m_1 g$) acts downward and normal force N exerted by the fixed support at the center acts upward. The tension T acts downward at the edge. The gravitational force ($m_1 g$) and the normal force (N) cancel each other. $m_1 g = N$

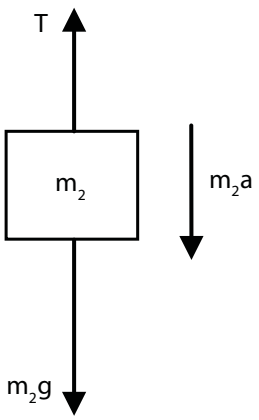
The tension T produces a torque (R T), which produces a rotational motion in the disc with angular acceleration, $\left(\alpha = \frac{a}{R}\right)$. Here, a is the linear acceleration of a point at the edge of the disc. If the moment of inertia of the disc is I and its radius of gyration is K, then

$$R\,T = I\alpha; \quad R\,T = \left(m_1 K^2\right)\frac{a}{R}$$

$$T = \left(m_1 K^2\right)\frac{a}{R^2}$$

FBD of the body:

Its gravitational force ($m_2 g$) acts downward and the tension T acts upward. As ($T < m_2 g$), there is a resultant force ($m_2 a$) acting on it downward.



$$m_2 g - T = m_2 a$$

Substituting for T from the equation for disc,

$$m_2 g - \left(m_1 K^2\right)\frac{a}{R^2} = m_2 a$$

$$m_2 g = \left(m_1 K^2\right)\frac{a}{R^2} + m_2 a$$

$$m_2 g = \left[\left(m_1 \frac{K^2}{R^2}\right) + m_2\right] a$$

$$a = \frac{m_2}{\left[\left(m_1 \frac{K^2}{R^2}\right) + m_2\right]} g$$

The expression $\left(\frac{K^2}{R^2}\right)$ for a disc rotating about an axis passing through the center and perpendicular to the plane is, $\frac{K^2}{R^2} = \frac{1}{2}$. (Ref Table 5.3) Now the expression for acceleration further simplifies as,

$$a = \frac{m_2}{\left[\left(\frac{m_1}{2}\right) + m_2\right]} g; \quad a = \frac{2m_2}{\left[m_1 + 2m_2\right]} g$$

substituting the values,

$$a = \frac{2 \times 0.1}{\left[0.5 + 0.2\right]} \times 10 = \frac{0.2}{0.7} \times 10$$

$$a = 2.857\ m\,s^{-2}$$

5.5.2 Conservation of Angular Momentum

When no external torque acts on the body, the net angular momentum of a rotating rigid body remains constant. This is known as law of conservation of angular momentum.